

H.W

Write a C++ code which includes a class student. find out the space occupied by the object of the class.

Algorithm

Efficiency

→ Although computer scientists + Brassard and Bratley coined the term algorithmics, which they defined as

"The systematic study of fundamental techniques used to design and analyse efficient algorithms"

$f(n)$ = efficiency

the order of growth

```
#include <iostream>
```

```
using namespace std;
```

```
class student
```

```
{
```

```
    int id; (4)
```

```
    string name; (32)
```

```
public:
```

```
    student()
```

```
    {
```

```
        cin >> name >> id;
```

```
    }
```

```
    void display()
```

```
    {
```

```
        cout << name << id;
```

```
    }
```

```
};
```

```
int main()
```

```
{
```

```
    student s;
```

```
    s.display();
```

```
    cout << sizeof(s);
```

```
}
```

int → 4 bytes

string → 32 bytes

36 bytes

padding

01 → 4 bytes

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Total: 45 bytes